

CARAVELAS, Brazil

WMO: 834970

Lat: 17.652S

Lon: 39.253W

Elev: 36

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 83-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	61.2	63.0	58.7	73.8	68.5	60.4	78.5	68.1	23.1	74.3	21.1	75.9	0.8	240	0.280

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.0	89.3	77.3	88.1	77.1	87.6	77.0	79.1	85.9	78.5	85.0	78.1	84.4	11.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.2	81.3	76.9	140.2	80.9	76.3	137.6	80.4	42.6	86.2	42.0	85.8	41.5	84.6	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.6	20.0	17.7	DB	57.9	94.3	1.8	3.3	56.6	96.7	55.5	98.6	54.5	100.5	53.2	102.9
			WB	55.9	83.0	2.7	2.2	54.0	84.5	52.4	85.8	51.0	87.0	49.0	88.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.6	79.9	80.5	80.2	78.7	76.3	73.5	72.3	72.1	73.7	75.7	77.6	78.6
	DBStd	3.70	1.60	1.47	1.63	1.90	2.32	2.27	2.28	2.37	2.58	2.63	2.60	2.23
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9698	928	853	937	861	815	706	691	684	711	796	828	888
	CDD65	4223	463	433	472	411	350	256	226	219	261	331	378	423
	CDH74	30901	4286	4225	4469	3321	2249	1214	911	918	1297	1973	2695	3343
	CDH80	7952	1331	1482	1512	993	516	165	83	59	132	311	551	817
(14) Wind	WSAvg	6.0	6.7	5.4	5.9	4.9	4.6	4.1	4.4	5.3	7.1	7.5	8.6	7.9
(15) Precipitation	PrecAvg	56.3	4.0	3.1	5.4	5.5	5.0	3.4	4.5	2.8	3.3	5.5	7.7	5.9
	PrecMax	90.2	15.7	12.6	21.1	12.3	13.1	9.7	11.3	8.0	10.5	15.9	18.3	15.9
	PrecMin	29.0	0.1	0.0	0.3	0.4	0.0	0.2	0.7	0.4	0.2	0.2	0.4	0.5
	PrecStd	13.1	3.7	2.8	4.1	2.9	2.8	2.0	2.6	1.7	2.3	3.7	4.4	3.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	90.0	90.0	89.4	87.8	84.6	83.5	83.1	84.1	86.3	87.1	88.2
		MCWB	77.8	77.8	77.2	77.2	77.6	74.9	73.7	73.3	74.0	75.5	77.1	77.4
	2%	DB	88.0	89.0	89.1	88.0	86.0	83.1	82.2	81.5	82.6	84.5	85.8	86.5
		MCWB	77.2	77.5	77.0	77.2	76.7	74.2	73.1	72.5	73.6	75.0	76.0	76.7
	5%	DB	87.0	88.0	88.0	87.0	84.5	82.1	80.8	80.4	81.6	82.8	84.4	85.7
		MCWB	76.9	77.1	76.9	77.1	76.0	73.9	72.6	72.0	73.1	74.3	75.6	76.4
	10%	DB	86.0	87.2	87.0	85.7	83.1	80.5	79.1	78.9	80.4	82.0	83.4	84.4
		MCWB	76.5	77.0	76.9	76.9	75.3	73.2	72.1	71.3	72.5	74.0	75.3	76.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.8	79.8	79.8	79.4	78.7	76.5	75.5	75.2	76.0	77.2	78.6	79.1
		MCDB	86.6	87.5	86.9	85.9	85.6	82.1	81.3	80.2	81.0	83.5	85.9	86.6
	2%	WB	78.7	78.9	78.9	78.3	77.5	75.4	74.3	73.8	74.9	76.1	77.3	77.8
		MCDB	85.1	85.9	85.2	84.8	83.7	81.0	79.9	79.1	79.9	82.1	83.2	84.0
	5%	WB	78.0	78.2	78.2	77.8	76.8	74.7	73.4	73.0	74.2	75.4	76.5	77.2
		MCDB	84.0	84.9	84.4	84.1	82.6	79.9	78.8	78.2	79.0	81.0	81.8	82.7
	10%	WB	77.3	77.6	77.7	77.2	76.1	73.9	72.6	72.1	73.5	74.7	76.0	76.5
		MCDB	83.1	84.2	83.8	83.5	81.5	78.8	77.6	77.0	78.1	79.8	81.1	81.6
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	14.0	13.9	13.6	13.9	14.3	14.0	13.9	12.9	11.4	10.9	11.3
		MCDBR	14.1	15.1	15.5	15.6	15.7	16.4	16.4	16.0	14.0	13.0	12.7	13.3
		MCWBR	4.8	4.9	5.0	5.9	6.8	7.9	8.1	7.6	5.9	5.3	4.9	4.8
	5% WB	MCDBR	13.0	14.0	13.7	14.6	14.6	14.7	15.0	14.9	12.7	11.5	11.4	12.1
		MCWBR	5.1	5.3	5.2	6.2	6.8	7.9	7.9	7.8	6.0	5.3	4.9	4.9
(40) Clear-Sky Solar Irradiance	taub		0.395	0.388	0.387	0.382	0.376	0.363	0.360	0.375	0.431	0.434	0.419	0.415
	taud		2.502	2.525	2.546	2.548	2.545	2.569	2.560	2.490	2.331	2.354	2.431	2.444
	Ebn at Noon		301	300	293	283	273	272	276	281	276	283	292	295
	Edh at Noon		36	36	34	31	30	28	29	33	41	42	39	39
(44) All-Sky Solar Radiation	RadAvg		2095	2045	1829	1574	1323	1230	1307	1519	1641	1780	1768	1957
	RadStd		181	135	141	107	66	77	69	73	136	233	205	193

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page